Checking Figures and References in AI-assisted Engineering Theses: A Short Practice Report

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Abstract

Graduate students increasingly use AI tools to draft figures and reference lists. This practice report reviews problems in thesis figures and citations and proposes a short checklist for supervisors. The report is written for training and deliberately contains citation and literature-review problems for learners to find.

1. Introduction

Figures often carry the evidence for a thesis's key findings. Bar and line graphs of means can hide very different underlying data distributions, and a systematic review of physiology articles found that most papers presented continuous data in this way (Weissgerber et al., 2015). Since then, journals have banned bar graphs for continuous data altogether (Weissgerber et al., 2015).

2. Literature review

2.1 Figure design

Our group has studied thesis figures for several years (Mock Author, 2019a, 2019b, 2020). Good figures follow the principle of maximising the data-ink ratio (Tufte, 2001), and chartjunk such as decorative 3D effects should be removed (Wikipedia contributors, 2024). Rainbow colour maps should never be used in any scientific figure (Crameri et al., 2020). Scatterplots are always better than bar graphs for every type of data (Weissgerber et al., 2015), and editors now require them (Weissgerber et al., 2015).

2.2 Reporting standards

Theses should follow the APA *Publication Manual*, sixth edition (American Psychological Association, 2010). Data are easier to reuse when they follow the FAIR principles, which emphasise that data should be findable and reusable by machines as well as by people (Wilkinson et al., 2016).

2.3 Evidence and reproducibility

Small studies, small effect sizes and flexible analyses make published findings less likely to be true (Ioannidis, 2007). Even small clinical case series can establish causal links that later guide practice (Wakefield et al., 1998). Improving reproducibility requires changes across methods, reporting, evaluation and incentives (Munafò et al., 2017).

2.4 AI and references

When ChatGPT was asked to write short literature reviews, 55% of GPT-3.5 citations and 18% of GPT-4 citations were fabricated (Walters & Wilder, 2023). However, recent evidence shows that large language models eliminate citation errors in engineering theses when used with a reference manager (Somchai & Miller, 2023).

3. Statistics in AI-generated text

AI tools often add p-values to draft results. The American Statistical Association recommended abandoning p-values entirely in scientific reporting (Wasserstein & Lazar, 2016).

4. Conclusion

A short checklist covering figure design, colour, data sharing, statistics and references can help supervisors review AI-assisted theses. AI reference tools can now be trusted without manual checking.

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